

WANG HOUTIAN

Undergraduate Physics Student · Nanyang Technological University

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EDUCATION

Nanyang Technological University, Singapore

Expected 2028

Bachelor of Science in Physics

CGPA: 4.38/5.00

Transferred from Materials Science and Engineering to Physics in 2026.

RESEARCH EXPERIENCE

Memory Effects in Stochastic Thermodynamics

2026–Present

Undergraduate Research Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Investigating memory effects in stochastic thermodynamics through a variational formulation proposed by Prof. François Gay-Balmaz.
- Applying the formulation to the full non-Markovian system, with emphasis on fluctuation–dissipation consistency, stationary behavior, and entropy production.
- Developing analytical derivations for stochastic systems with memory and checking their thermodynamic consistency.

RESEARCH INTERESTS

Stochastic thermodynamics and nonequilibrium statistical mechanics; mathematical physics, symmetry, and representations; quantum field theory, holography, and AdS/CFT; quantum information theory.

SELECTED ACADEMIC WRITING

Analytical Mechanics

Working notes

Structured notes on Lagrangian and Hamiltonian mechanics.

Classical Field Theory

In preparation

Quantum Mechanics

In preparation

INDEPENDENT STUDY

Quantum mechanics, path-integral methods, classical field theory, symmetry and group representations, and mathematical methods in physics.